

MA1001 — Calculus for Science Part 1

Winter 2024 Examination · Paper 1 · Full Marking Scheme

Module	MA1001 — Calculus for Science Part 1
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Institution	University College Cork
Session	Winter 2024, Paper 1
Duration	1.5 hours (+ 15 minutes reading time)
Structure	Answer all 6 questions ; all questions carry equal marks (10 each in this scheme, 60 in total).
Materials	No calculators or electronic devices. Formula sheet supplied with the paper (calculus, trigonometry, length/area/volume).

How to use this scheme. Every answer is worked out in small numbered steps, with a “Rules you need” box listing the rules used. Nothing is skipped, so you can see where every number comes from. Read the answer first, then follow the steps.

No calculators. Every answer here is exact (for example $\ln 2$, $\pi + 1$, $-15/2$), and all the arithmetic is shown by hand. Rough values like “ $\ln 2$ is about 0.7” are only there to help you plot a sketch.

Coverage note. This scheme works through **all six questions and every part** in full. There are no either/or options on this paper.

Source basis. Every answer is built **only** from the MA1001 lecture notes and example/tutorial sheets: logarithms, graph sketching and transformations, derivative rules (product, quotient, chain), implicit differentiation, increasing/decreasing functions and the first derivative test, asymptotes, and rates of change. No textbook or outside material is used.

Mark splits. The paper gives equal marks to each question but doesn't split them into parts. The part and step marks shown are Study-Uni's suggested allocation, 10 per question.

Own-figure rule. Where a later step uses an earlier result, a correct method applied to a wrong earlier value still earns the method marks.

Study-Uni

Question 1

Piecewise functions, logarithms, graph transformations — 10 marks

Given: $f(x) = x^2$ if $x < -1$; 0 if $-1 \leq x < 1$; $\ln x$ if $x \geq 1$

1(a)

[2 marks]

What are the values $f(-1)$ and $f(1)$?

ANSWER

$$f(-1) = 0 \quad \text{and} \quad f(1) = \ln 1 = 0$$

WORKING — STEP BY STEP

1 How to read a piecewise function

f has three different rules. For any number x , check which condition x satisfies and use **only** that rule.

Rule 1: $x < -1$ | Rule 2: $-1 \leq x < 1$ | Rule 3: $x \geq 1$

2 Find $f(-1)$: which rule does $x = -1$ belong to?

Rule 1 needs $-1 < -1$. That's false (a number is not less than itself).

Rule 2 needs $-1 \leq -1$ **and** $-1 < 1$. Both are true, so use Rule 2, which always gives 0:

$$f(-1) = 0$$

(Be careful: it is **not** $(-1)^2 = 1$, because Rule 1 does not include $x = -1$.)

3 Find $f(1)$: which rule does $x = 1$ belong to?

Rule 2 needs $1 < 1$: false. Rule 3 needs $1 \geq 1$: true ($\geq$ allows equal). So use $\ln x$:

$$f(1) = \ln 1$$

$\ln 1 = 0$, because $e^0 = 1$ (the log undoes the exponential: $\ln(e^x) = x$, so $\ln(e^0) = 0$).

$$f(1) = 0$$

EXAM TECHNIQUE

The end points are the trap. $\leq$ and $\geq$ include the number itself; $<$ and $>$ don't. So -1 goes with $-1 \leq x < 1$, and 1 goes with $x \geq 1$.

Marks

Awarded for

1 $f(-1) = 0$ (correct rule chosen).

1 $f(1) = \ln 1 = 0$.

1(b)**[2 marks]**Is f decreasing on $(-\infty, -1)$? Justify your answer.**ANSWER****Yes. On $(-\infty, -1)$, $f'(x) = 2x$, which is negative, so f is decreasing.****RULES YOU NEED**

- f is decreasing where $f'(x) < 0$.
- Power rule: the derivative of x^n is $n \cdot x^{n-1}$.

WORKING — STEP BY STEP**1 Which rule of f do we use here?** $(-\infty, -1)$ means every number less than -1 (e.g. $-2, -5, -100$). Rule 1 applies:

$$f(x) = x^2$$

2 The key idea: the sign of the derivativeA function is **decreasing** where $f'(x) < 0$ and **increasing** where $f'(x) > 0$. So differentiate and check the sign.**3 Differentiate x^2**

Use the power rule: bring the power down in front and take 1 off the power.

$$f'(x) = 2x^{2-1} = 2x$$

4 Is $f'(x)$ negative when $x < -1$?Every x in the interval is negative, and $2 \times$ (a negative number) is negative. For example:

$$x = -3: f'(-3) = 2 \times (-3) = -6 \quad (\text{negative})$$

$$f'(x) = 2x < 0 \text{ for every } x \text{ in } (-\infty, -1)$$

5 Conclusion (and a quick check with numbers)Since $f'(x) < 0$ on the whole interval, **yes, f is decreasing on $(-\infty, -1)$.**Check: as x goes up from -3 to -2 , f goes down:

$$f(-3) = (-3)^2 = 9 \rightarrow f(-2) = (-2)^2 = 4$$

Marks	Awarded for
1	$f'(x) = 2x$ on this interval.
1	$2x < 0$ when $x < -1$, so yes, f is decreasing.

1(c)**[3 marks]**Sketch the graph of $y = f(x)$ for $-2 < x < 2$.**ANSWER**

See the sketch: part of the parabola $y = x^2$ falling from $(-2, 4)$ towards $(-1, 1)$, a jump down to the x-axis, flat along the x-axis from -1 to 1 , then the curve $y = \ln x$ rising from $(1, 0)$ towards $(2, \ln 2)$.

WORKING — STEP BY STEP**1 Split $-2 < x < 2$ into the three rules**

$$-2 < x < -1: y = x^2 \quad | \quad -1 \leq x < 1: y = 0 \quad | \quad 1 \leq x < 2: y = \ln x$$

2 Left piece: table of values for $y = x^2$

x	-2	-1.5	-1
$y = x^2$	4	2.25	1

For example $(-1.5)^2 = (-1.5) \times (-1.5) = 2.25$. The y values go **down** from 4 to 1, matching part (b).

Rule 1 does not include $x = -1$, so draw an **open dot** at $(-1, 1)$. The end $x = -2$ is not included either ($-2 < x$), so an **open dot** at $(-2, 4)$.

3 Middle piece: $y = 0$ for $-1 \leq x < 1$

The graph sits on the x-axis from $x = -1$ to $x = 1$.

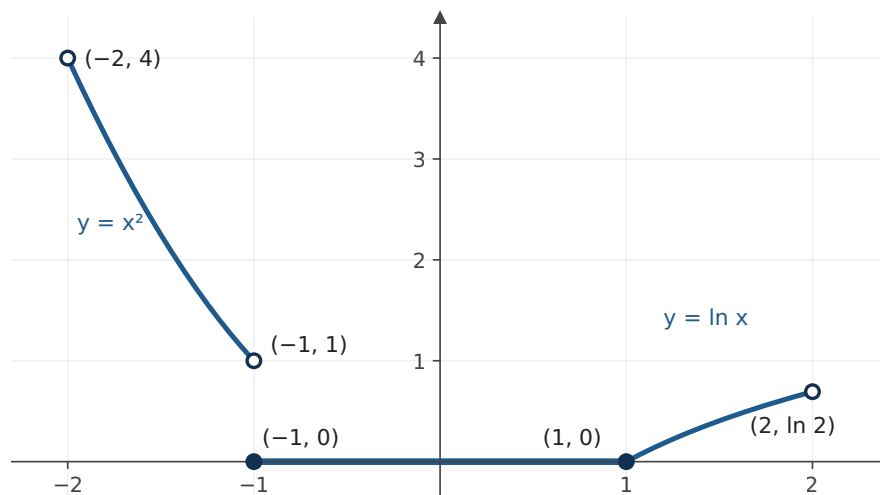
$\leq$ includes -1 , so a **filled dot** at $(-1, 0)$. So at $x = -1$ the graph **jumps** from the open dot at height 1 down to the filled dot at 0.

4 Right piece: $y = \ln x$ for $1 \leq x < 2$

Key facts about $\ln x$ (no calculator needed): $\ln 1 = 0$, $\ln x$ is increasing, and $\ln 2$ is a bit less than 1 (about 0.7, since $e \approx 2.7$ is bigger than 2).

At $x = 1$ the middle piece reaches height 0 and $\ln 1 = 0$ too, so the graph **joins up** with no jump: a **filled dot** at $(1, 0)$.

It then rises to an **open dot** at $(2, \ln 2)$, as $x = 2$ is not included.



Marks	Awarded for
1	Parabola piece falling from $(-2, 4)$ to an open dot at $(-1, 1)$.
1	Filled dot at $(-1, 0)$ and flat line on the x-axis up to $x = 1$.
1	$\ln x$ curve starting at $(1, 0)$ and rising towards $(2, \ln 2)$.

1(d)**[3 marks]**Sketch the graph of $g(x) = f(-x) + 1$ for $-2 < x < 2$.**ANSWER**

$g(x) = 1 + \ln(-x)$ for $x \leq -1$; 1 for $-1 < x \leq 1$; $1 + x^2$ for $x > 1$ (see sketch).

WORKING — STEP BY STEP**1 What the two changes do to a graph**

$f(-x)$: put $-x$ in place of x . This flips the graph left-to-right (a mirror image in the y -axis).

$+ 1$: adds 1 to every y value, so the whole graph moves up by 1.

2 Work out $f(-x)$: replace x by $-x$ in each rule

Each condition changes too. Multiplying an inequality by -1 **flips its direction** (e.g. $-x < -1$ becomes $x > 1$).

$$\text{Rule 1: } -x < -1 \Leftrightarrow x > 1 \text{ gives } (-x)^2 = x^2$$

$$\text{Rule 2: } -1 \leq -x < 1 \Leftrightarrow -1 < x \leq 1 \text{ gives } 0$$

$$\text{Rule 3: } -x \geq 1 \Leftrightarrow x \leq -1 \text{ gives } \ln(-x)$$

($\ln(-x)$ makes sense here because $x \leq -1$ means $-x$ is positive.)

3 Add 1 to each rule to get $g(x)$

$$g(x) = 1 + \ln(-x) \text{ if } x \leq -1$$

$$g(x) = 1 \text{ if } -1 < x \leq 1$$

$$g(x) = 1 + x^2 \text{ if } x > 1$$

4 Table of values

x	-2	-1.5	-1	0	1	1.5	2
$g(x)$	$1 + \ln 2$	$1 + \ln 1.5$	1	1	1	3.25	5

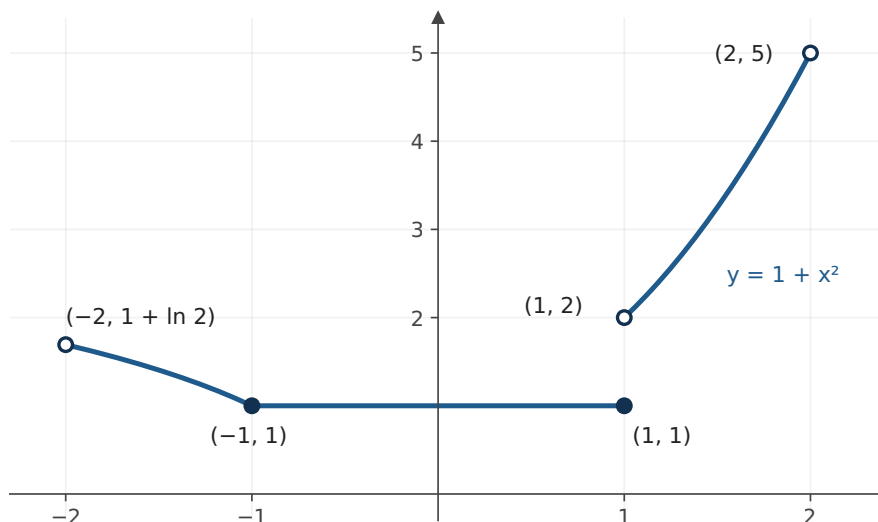
For example $g(-1) = 1 + \ln(1) = 1 + 0 = 1$, and $g(1.5) = 1 + 1.5^2 = 1 + 2.25 = 3.25$. ($1 + \ln 2$ is about 1.7.)

5 Dots at the break points

At $x = -1$: the first rule has $\leq$, so a **filled dot** at $(-1, 1)$. The flat part also sits at height 1, so the graph joins up here.

At $x = 1$: the middle rule includes 1 ($\leq$), so a **filled dot** at $(1, 1)$. The right rule has $>$, so it starts with an **open dot** at $(1, 1 + 1^2) = (1, 2)$. The graph **jumps up** here.

The ends $x = \pm 2$ are not included, so draw **open dots** at $(-2, 1 + \ln 2)$ and $(2, 5)$.

**EXAM TECHNIQUE**

Do the mirror flip first, then move up by 1. The jump that was at $x = -1$ in part (c) is now at $x = 1$.

Marks	Awarded for
1	$f(-x)$ worked out correctly (mirror image in the y-axis).
1	Shifted up by 1: falling log curve down to the filled dot $(-1, 1)$, then flat at $y = 1$ up to $x = 1$.
1	Jump at $x = 1$ to an open dot at $(1, 2)$, then the curve $1 + x^2$ rising towards $(2, 5)$.

Question 1 Total: 10 Marks

Question 2

Logarithm rules, solving equations — 10 marks

2

[10 marks]

Solve the equation $\log_2(x + 1) = 3 - \log_2(x - 1)$.

ANSWER

$x = 3$ ($x = -3$ also comes out of the algebra but must be rejected)

RULES YOU NEED

- $\log A + \log B = \log(A \times B)$
- $\log_2 y = 3$ means $y = 2^3 = 8$
- You can only take the log of a positive number.

WORKING — STEP BY STEP

1 Find which x values are allowed

You can only take the log of a **positive** number. So both inputs must be positive:

$$x + 1 > 0 \Rightarrow x > -1 \quad \text{and} \quad x - 1 > 0 \Rightarrow x > 1$$

Both must be true, so:

$$x > 1$$

2 Get both logs on the same side

Add $\log_2(x - 1)$ to both sides:

$$\log_2(x + 1) + \log_2(x - 1) = 3$$

3 Combine the two logs into one

Use the rule: $\log A + \log B = \log(A \times B)$.

$$\log_2[(x + 1)(x - 1)] = 3$$

4 Tidy the bracket

Multiply out $(x + 1)(x - 1) = x^2 - x + x - 1 = x^2 - 1$ (the difference of two squares):

$$\log_2(x^2 - 1) = 3$$

5 Remove the log

$\log_2(\text{something}) = 3$ means the something equals 2^3 , and $2^3 = 2 \times 2 \times 2 = 8$:

$$x^2 - 1 = 8$$

6 Solve for x

Add 1 to both sides, then take the square root of both sides. Remember a square root gives **two** answers, + and −:

$$x^2 = 9 \Rightarrow x = 3 \text{ or } x = -3$$

7 Check both answers against Step 1

x = −3: $x - 1 = -4$, which is negative, so $\log_2(-4)$ doesn't exist. **Reject x = −3.**

x = 3: $x + 1 = 4$ and $x - 1 = 2$, both positive ✓. Check it in the original equation (no calculator needed, as $4 = 2^2$ and $2 = 2^1$):

$$\text{Left: } \log_2 4 = 2 \quad \text{Right: } 3 - \log_2 2 = 3 - 1 = 2 \quad \checkmark$$

So the only solution is **x = 3**.

EXAM TECHNIQUE

Always check every answer in the original equation. Combining the logs hides the rule that each log needs a positive input, so a fake answer can appear (here $x = -3$).

Marks	Awarded for
2	Conditions $x + 1 > 0$ and $x - 1 > 0$ stated (or both roots checked in the original equation).
2	Logs combined using $\log A + \log B = \log(AB)$ (or writing $3 = \log_2 8$).
2	$x^2 - 1 = 8$.
2	$x = \pm 3$.
2	$x = -3$ rejected with a reason; final answer $x = 3$.

Question 2 Total: 10 Marks

Question 3

Chain rule, tangent lines — 10 marks

3

[10 marks]

Find the equation of the tangent line to the curve $y = e^{\sin x}$ at the point where $x = \pi$.

ANSWER

$$y = -x + \pi + 1 \quad (\text{the slope is } -1)$$

RULES YOU NEED

- Tangent line: $y - b = m(x - a)$, where $m = dy/dx$ at the point (a, b) .
- Chain rule: $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$.
- The derivative of e^u is e^u , and of $\sin x$ is $\cos x$. Trig table: $\sin \pi = 0$, $\cos \pi = -1$.

WORKING — STEP BY STEP

1 What we need for a tangent line

A straight line needs **a point** (a, b) and **a slope** m . Then use:

$$y - b = m(x - a)$$

The slope of the tangent is the derivative dy/dx at that point.

2 Find the point (put $x = \pi$ into the curve)

Work from the inside out. From the trig table, $\sin \pi = 0$. Anything to the power 0 is 1:

$$y = e^{\sin \pi} = e^0 = 1 \quad \Rightarrow \quad \text{point } (\pi, 1)$$

3 Differentiate using the chain rule

$y = e^{\sin x}$ is a function inside a function: $e^{(\quad)}$ on the outside, $\sin x$ on the inside. Let $u = \sin x$.

$$y = e^u \Rightarrow \frac{dy}{du} = e^u \quad u = \sin x \Rightarrow \frac{du}{dx} = \cos x$$

Multiply them together, then put $u = \sin x$ back in:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = e^{\sin x} \times \cos x$$

4 Find the slope at $x = \pi$

From the trig table: $\sin \pi = 0$ and $\cos \pi = -1$.

$$m = e^{\sin \pi} \times \cos \pi = e^0 \times (-1) = 1 \times (-1) = -1$$

5 Write down the equation of the line

Put $a = \pi$, $b = 1$ and $m = -1$ into $y - b = m(x - a)$:

$$y - 1 = -1 \times (x - \pi) = -x + \pi$$

Add 1 to both sides:

$$y = -x + \pi + 1$$

EXAM TECHNIQUE

Leave π in the answer: no calculators are allowed, and $y = -x + \pi + 1$ is exact. Also note $e^0 = 1$, not 0.

Marks	Awarded for
2	Point $(\pi, 1)$.
3	$dy/dx = e^{\sin x} \cos x$ using the chain rule.
2	Slope $m = -1$ at $x = \pi$.
3	Tangent line $y = -x + \pi + 1$ (or $y - 1 = -(x - \pi)$).

Question 3 Total: 10 Marks

Question 4

Related rates of change, implicit differentiation — 10 marks

4

[10 marks]

You run around a circular track of radius 100 meters centered at the origin of coordinates. At the moment when you reach the point $(x, y) = (60, 80)$, your x -coordinate is increasing at a rate of 10 meters per second. At what rate is your y -coordinate increasing?

ANSWER

$dy/dt = -7.5$ m/s: your y -coordinate is actually decreasing at 7.5 m/s

RULES YOU NEED

- Circle with centre $(0, 0)$ and radius r : $x^2 + y^2 = r^2$.
- Chain rule for a quantity that depends on time: $d/dt(y^2) = 2y \cdot dy/dt$.

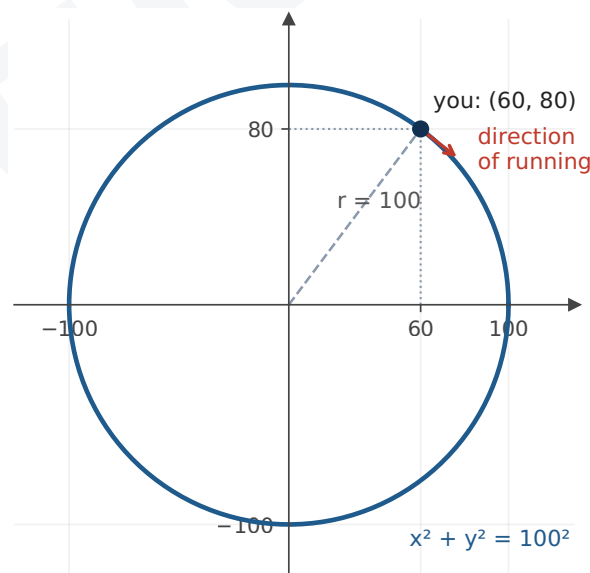
WORKING — STEP BY STEP

1 Write down the equation of the track

Every point on a circle with centre $(0, 0)$ and radius r satisfies $x^2 + y^2 = r^2$. Here $r = 100$:

$$x^2 + y^2 = 100^2 = 10\,000$$

Check the point is on it: $60^2 + 80^2 = 3600 + 6400 = 10\,000$ ✓



2 Write down what we know and what we want

Both x and y change as time t passes. We know

$$\frac{dx}{dt} = 10 \text{ m/s} \quad \text{when} \quad x = 60, y = 80$$

and we want $\frac{dy}{dt}$ at that moment.

3 Differentiate both sides with respect to time t

x and y are both functions of t , so use the chain rule on each square (as in implicit differentiation):

$$\frac{d}{dt}(x^2) = 2x \frac{dx}{dt} \quad \frac{d}{dt}(y^2) = 2y \frac{dy}{dt} \quad \frac{d}{dt}(10\,000) = 0$$

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

4 Put in the numbers

$x = 60$, $y = 80$ and $dx/dt = 10$:

$$2(60)(10) + 2(80) \frac{dy}{dt} = 0 \Rightarrow 1200 + 160 \frac{dy}{dt} = 0$$

5 Solve for dy/dt

Subtract 1200 from both sides, then divide by 160:

$$160 \frac{dy}{dt} = -1200 \Rightarrow \frac{dy}{dt} = \frac{-1200}{160} = -\frac{15}{2} = -7.5 \text{ m/s}$$

($1200 \div 160$: divide top and bottom by 80 to get $15 \div 2 = 7.5$.)

6 What the answer means

The rate is negative, so your y -coordinate is **decreasing** at 7.5 m/s (it is “increasing” at a rate of -7.5 m/s). This makes sense on the diagram: at $(60, 80)$, moving to the right on the top of the circle takes you downwards.

EXAM TECHNIQUE

Don't drop the minus sign. The question asks how fast y is increasing, and the honest answer is -7.5 m/s, i.e. y is going down.

Marks	Awarded for
2	Equation of the track $x^2 + y^2 = 100^2$.
3	Differentiating with respect to t : $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$.
2	Substituting $x = 60$, $y = 80$, $dx/dt = 10$.
3	$dy/dt = -7.5$ m/s (y decreasing at 7.5 m/s).

Question 4 Total: 10 Marks

Question 5

Critical points, increasing and decreasing functions, curve sketching — 10 marks

Given: $f(x) = \frac{x}{x^2 + 1}$

5 · Critical points

[4 marks]

Find all of the critical points of the function $f(x) = \frac{x}{x^2 + 1}$.

ANSWER

Critical points at $x = -1$ and $x = 1$, i.e. the points $(-1, -\frac{1}{2})$ and $(1, \frac{1}{2})$

RULES YOU NEED

- Critical point: where $f'(x) = 0$ (or f' does not exist).
- Quotient rule: if $f = u/v$ then $f' = (u' \cdot v - u \cdot v') / v^2$.
- Tidy the top, but never multiply out the bottom.

WORKING — STEP BY STEP

1 What a critical point is

A critical point is where the slope is zero: $f'(x) = 0$ (or where f' doesn't exist). So differentiate, then solve $f'(x) = 0$.

2 Set up the quotient rule

f is a fraction, so use the quotient rule. Name the top u and the bottom v , and differentiate each:

$$u = x \Rightarrow u' = 1 \quad v = x^2 + 1 \Rightarrow v' = 2x$$

$$f'(x) = \frac{u' \cdot v - u \cdot v'}{v^2}$$

3 Substitute into the formula

$$f'(x) = \frac{(1)(x^2 + 1) - (x)(2x)}{(x^2 + 1)^2}$$

4 Tidy up the top (never multiply out the bottom)

$$(1)(x^2 + 1) - (x)(2x) = x^2 + 1 - 2x^2 = 1 - x^2$$

$$f'(x) = \frac{1 - x^2}{(x^2 + 1)^2} = \frac{(1 - x)(1 + x)}{(x^2 + 1)^2}$$

$(1 - x^2 = (1 - x)(1 + x))$ is the difference of two squares.)

5 Solve $f'(x) = 0$

The bottom $(x^2 + 1)^2$ is always positive, never 0, so $f'(x)$ exists everywhere. A fraction is 0 only when its top is 0:

$$1 - x = 0 \Rightarrow x = 1 \quad 1 + x = 0 \Rightarrow x = -1$$

6 Find the y-value at each critical point

Put each x back into the original $f(x)$:

$$f(1) = \frac{1}{1^2 + 1} = \frac{1}{2} \quad f(-1) = \frac{-1}{(-1)^2 + 1} = \frac{-1}{2} = -\frac{1}{2}$$

Marks	Awarded for
2	$f'(x) = (1 - x^2)/(x^2 + 1)^2$ using the quotient rule.
1	$f'(x) = 0$ when $1 - x^2 = 0$.
1	$x = \pm 1$ with their y-values $\pm \frac{1}{2}$.

5 · Increasing / decreasing**[3 marks]**

Find the intervals on which the function is increasing, and those on which it is decreasing.

ANSWER**Decreasing on $(-\infty, -1]$ and $[1, \infty)$; increasing on $[-1, 1]$. Local minimum at $x = -1$, local maximum at $x = 1$.****WORKING — STEP BY STEP****1 The rule**

f is **increasing** where $f'(x) > 0$ and **decreasing** where $f'(x) < 0$. The bottom of f' is always positive, so only the top, $1 - x^2$, decides the sign.

2 Mark the split points and choose test points

The **split points** are the critical points $x = -1$ and $x = 1$. They cut the number line into three pieces, and f' can only change sign at them. Pick one **test point** in each piece:

$$(-\infty, -1): \text{test } x = -2 \quad (-1, 1): \text{test } x = 0 \quad (1, \infty): \text{test } x = 2$$

3 Work out the sign of f' at each test point

$$x = -2: 1 - (-2)^2 = 1 - 4 = -3 \rightarrow f'(-2) = -3/25 \text{ negative}$$

$$x = 0: 1 - 0^2 = 1 \rightarrow f'(0) = 1/1 = 1 \text{ positive}$$

$$x = 2: 1 - 2^2 = -3 \rightarrow f'(2) = -3/25 \text{ negative}$$

(The bottom at $x = \pm 2$ is $(4 + 1)^2 = 25$.)

4 Summary table

interval	$(-\infty, -1)$	$(-1, 1)$	$(1, \infty)$
sign of $f'(x)$	–	+	–
f is	decreasing ↘	increasing ↗	decreasing ↘

5 Classify each critical point (first derivative test)

At $x = -1$ the graph goes down then up: a **valley**, so a local minimum, value $-\frac{1}{2}$.

At $x = 1$ it goes up then down: a **hill**, so a local maximum, value $\frac{1}{2}$.

Marks Awarded for

2 Correct sign of f' on each of the three intervals.

1 Intervals stated and critical points classified.

5 · Sketch**[3 marks]**Use all of this information to give a sketch of the graph of $y = f(x)$.**ANSWER**

A valley at $(-1, -\frac{1}{2})$, passing through the origin, a hill at $(1, \frac{1}{2})$, with both ends flattening out towards $y = 0$ (see the sketch).

WORKING — STEP BY STEP**1 Plot the critical points and the origin**

Valley (local minimum) at $(-1, -\frac{1}{2})$. Hill (local maximum) at $(1, \frac{1}{2})$. Also $f(0) = 0/1 = 0$, so the graph passes through $(0, 0)$.

2 Check for symmetry

Put $-x$ into f :

$$f(-x) = \frac{-x}{(-x)^2 + 1} = -\frac{x}{x^2 + 1} = -f(x)$$

So f is an **odd** function: the graph looks the same after turning it 180° around the origin. The valley at $(-1, -\frac{1}{2})$ mirrors the hill at $(1, \frac{1}{2})$.

3 Where is the graph above or below the x-axis?

The bottom $x^2 + 1$ is always positive, so $f(x)$ has the same sign as x : **above** the x-axis when $x > 0$ and **below** it when $x < 0$.

4 What happens far out (x very large)?

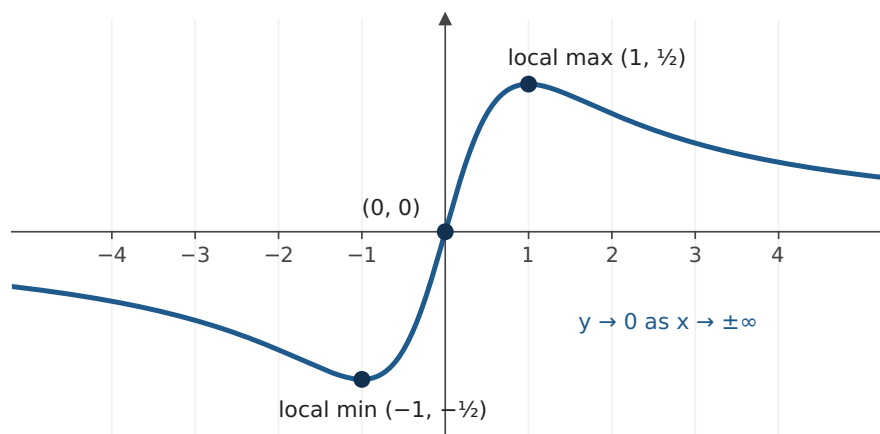
Divide the top and bottom by x^2 (the biggest power on the bottom):

$$f(x) = \frac{x / x^2}{x^2/x^2 + 1/x^2} = \frac{1/x}{1 + 1/x^2} \rightarrow \frac{0}{1 + 0} = 0$$

So $y = 0$ (the x-axis) is a **horizontal asymptote**: the graph gets closer and closer to it at both ends. Check: $f(10) = 10 \div 101$, just under 0.1.

5 Join it all up

Coming in from the left just below the x-axis, go down to the valley at $(-1, -\frac{1}{2})$, rise through the origin to the hill at $(1, \frac{1}{2})$, then fall back towards the x-axis, staying above it.



Marks	Awarded for
1	Odd symmetry, sign of f , or the horizontal asymptote $y = 0$ used.
2	Sketch with minimum at $(-1, -\frac{1}{2})$, maximum at $(1, \frac{1}{2})$, through the origin, rising and falling in the right places.
Question 5 Total: 10 Marks	

Question 6

Checking solutions of a differential equation — 10 marks

Given: $\frac{df}{dx} = -\frac{f}{x} + 3x$. Overall answer: (b) $f(x) = x^2$ and (c) $f(x) = x^2 + 1/x$ both satisfy it; (a) and (d) do not.

6(a)

[2 marks]

Does $f(x) = x$ satisfy $\frac{df}{dx} = -\frac{f}{x} + 3x$?

ANSWER

No

RULES YOU NEED

- Work out the **left side**, df/dx (the derivative of f).
- Work out the **right side**, $-f/x + 3x$ (divide f by x , put a minus in front, and add $3x$).
- Same for **every** $x \rightarrow$ yes. Different for even **one** $x \rightarrow$ no.

WORKING — STEP BY STEP

1 Left side: differentiate x

$$\frac{df}{dx} = 1$$

2 Right side: put $f = x$ in

$$x \div x = 1:$$

$$-\frac{x}{x} + 3x = -1 + 3x$$

3 Compare using one value of x

Try $x = 1$:

$$\text{Left: } 1 \quad \text{Right: } -1 + 3(1) = 2$$

1 is not equal to 2, so $f(x) = x$ does **not** satisfy the equation.

Marks

Awarded for

1 $df/dx = 1$ and the right side $-1 + 3x$.

1 Shown not equal (e.g. at $x = 1$): No.

6(b)**[3 marks]**Does $f(x) = x^2$ satisfy $\frac{df}{dx} = -\frac{f}{x} + 3x$?**ANSWER****Yes****WORKING — STEP BY STEP****1 Left side: differentiate x^2**

Power rule:

$$\frac{df}{dx} = 2x$$

2 Right side: put $f = x^2$ in $x^2 \div x = x$ (cancel one x from the top and bottom):

$$-\frac{x^2}{x} + 3x = -x + 3x = 2x$$

3 CompareLeft: $2x$ Right: $2x$ They are exactly the same for every x ($x \neq 0$, where the equation itself isn't defined). So **yes**.

Marks	Awarded for
1	$df/dx = 2x$.
1	Right side simplified to $2x$.
1	Equal for all x : Yes.

6(c)**[3 marks]**

Does $f(x) = x^2 + \frac{1}{x}$ satisfy $\frac{df}{dx} = -\frac{f}{x} + 3x$?

ANSWER**Yes****RULES YOU NEED**

- $1/x = x^{-1}$, and the power rule works for negative powers: the derivative of x^{-1} is $-x^{-2} = -1/x^2$.

WORKING — STEP BY STEP**1 Left side: differentiate $x^2 + 1/x$**

Write $1/x$ as a power: $1/x = x^{-1}$. Then use the power rule on each term:

$$\frac{df}{dx} = 2x + (-1)x^{-2} = 2x - \frac{1}{x^2}$$

2 Right side: put $f = x^2 + 1/x$ in

Divide **each** term of f by x : $x^2 \div x = x$ and $(1/x) \div x = 1/x^2$:

$$\frac{f}{x} = x + \frac{1}{x^2}$$

$$-\frac{f}{x} + 3x = -x - \frac{1}{x^2} + 3x = 2x - \frac{1}{x^2}$$

3 Compare

$$\text{Left: } 2x - \frac{1}{x^2} \quad \text{Right: } 2x - \frac{1}{x^2}$$

They are exactly the same for every $x \neq 0$. So **yes**.

Quick check at $x = 1$: left = $2 - 1 = 1$; right: $f(1) = 1 + 1 = 2$, so $-2/1 + 3 = 1$ ✓

Marks	Awarded for
1	$df/dx = 2x - 1/x^2$ (writing $1/x = x^{-1}$).
1	Right side simplified to $2x - 1/x^2$.
1	Equal for all $x \neq 0$: Yes.

6(d)**[2 marks]**

Does $f(x) = \cos x$ satisfy $\frac{df}{dx} = -\frac{f}{x} + 3x$?

ANSWER**No****WORKING — STEP BY STEP****1 Left side: differentiate $\cos x$**

$$\frac{df}{dx} = -\sin x$$

2 Right side: put $f = \cos x$ in

$$-\frac{\cos x}{x} + 3x$$

3 Compare using one value of x

Try $x = \pi$, where $\sin \pi = 0$ and $\cos \pi = -1$ (trig table):

$$\text{Left: } -\sin \pi = 0 \quad \text{Right: } -\frac{(-1)}{\pi} + 3\pi = \frac{1}{\pi} + 3\pi$$

The right side is two positive numbers added together, so it is definitely bigger than 0. The sides are not equal, so the answer is **no**.

EXAM TECHNIQUE

The equation involves f/x , so it only makes sense for $x \neq 0$. To say “yes”, show the two sides match for every such x , as in (b) and (c). Remember that more than one option can be correct.

Marks	Awarded for
1	$df/dx = -\sin x$ and the right side $-\cos x / x + 3x$.
1	Shown not equal (e.g. at $x = \pi$): No.

Question 6 Total: 10 Marks